



Birzeit University
Faculty of Engineering & Technology
Electrical & Computer Engineering Department
ENCS5323 - Wireless and Mobile Networks

Project Report

**Live Wi-Fi Channel Occupancy Sensing and Dynamic Channel-Hopping
Interference Generation with SDR**

Group Members:

Ruba Aldaghamin - 1220925

Alaa Sawalma - 1213536

Zainab Murad - 1211020

Instructor: Dr. Mohammad K. Jubran

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1. Introduction and Objectives

The 2.4 GHz band is open for anyone to use, so Wi-Fi, Bluetooth and many other devices all share it. This makes it crowded, and signals often overlap. Because of this it is useful to know which channels are busy, and to see what happens to a working link when another signal lands on its channel.

This project studies that with the ADALM-PLUTO radio. Part 1 senses the whole 2.4 GHz band, shows the busy Wi-Fi channels, and tracks how they change over time. Part 2 makes a Wi-Fi-like signal, senses the band, and moves to the clearest channel. We then measured what that signal does to a real Wi-Fi connection.

We had one PLUTO, which the course allows. So we ran the two parts one at a time instead of together. Running both on one radio is not useful, because the transmitter would drown out its own receiver. Every result below was measured on real hardware.

2. System Setup

The radio is an ADALM-PLUTO Rev.B. We tuned it across the 2.4 GHz band at a sample rate of 20 MSPS, which is the widest view it gives at once. We used a fixed (manual) receive gain so that every measurement uses the same reference. Automatic gain would change the scale on each capture and make the measurements impossible to compare.

The radio only sees 20 MHz at a time, but the band is 100 MHz. So we tuned it to several centre frequencies and joined the pieces together. From each piece we kept only the flat middle 75%, because the edges drop off on the radio's filter and would look falsely empty. We also threw away the first capture after each tune, since it still holds samples from the old frequency. The final spectrum has a 19.5 kHz resolution (5120 points across the band).

Our signal uses the real 802.11a/g OFDM structure: a 64-point inverse FFT, sub-carriers -26 to +26 with the middle one turned off and four pilot tones, and a 16-sample guard (the cyclic prefix). At 20 MSPS the sub-carriers are 312.5 kHz apart and each symbol is 4 microseconds long. This is what makes the signal look like Wi-Fi to the sensor instead of a plain tone. We change the width by changing the sample rate, and a short gap after each burst makes it bursty like real traffic.

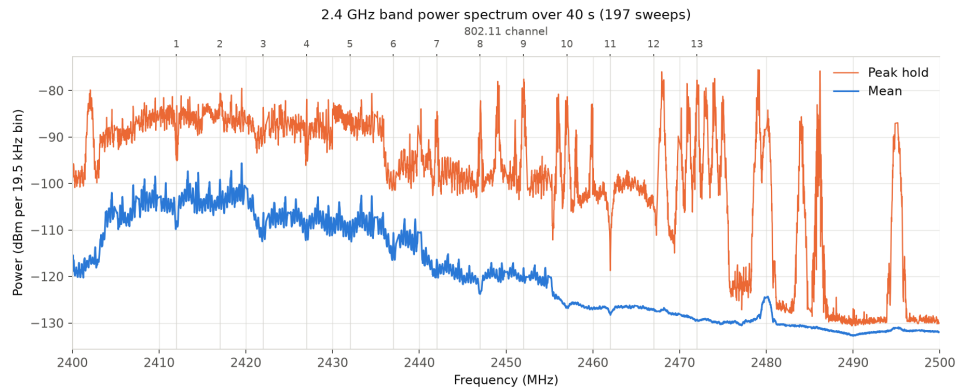
Table 1: Main experimental parameters.

Parameter	Value
SDR platform	ADALM-PLUTO Rev.B (AD9363A)
Sensing range	2400 - 2500 MHz
Sample rate	20 MSPS
Resolution bandwidth	19.5 kHz
Wi-Fi channel markers	Channels 1-13, 5 MHz spacing
Generated signal	802.11a/g OFDM, 20 MHz channel
Interference bandwidths	10 MHz, 20 MHz
Dwell per configuration	15 - 30 s
Receiver metric	Ping RTT and packet loss to router
Environment	Controlled indoor, single unit

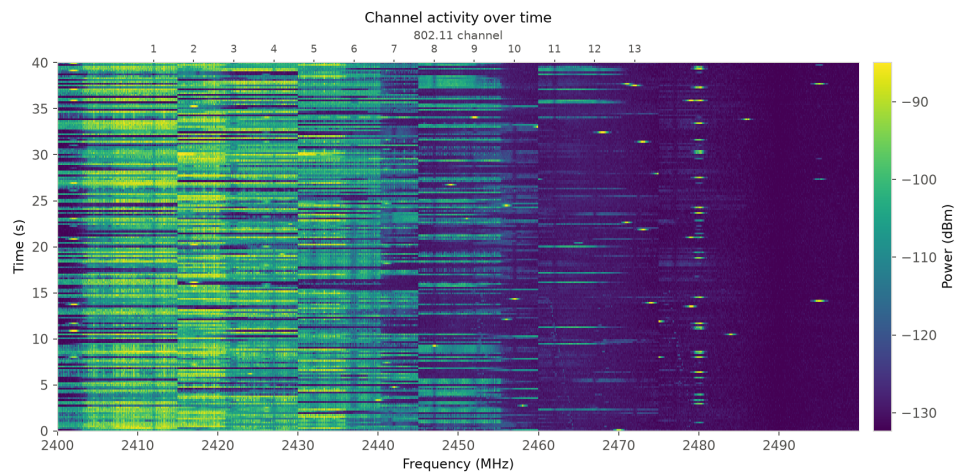
3. Part 1 - Spectrum Sensing

We ran the sensor for 40 seconds, which gave 197 full sweeps of the band. Figure 1 shows the power across the whole band with the Wi-Fi channels marked. The mean line shows the steady level and the peak line shows the bursts. The lower channels are clearly busy and the top of the band is almost empty.

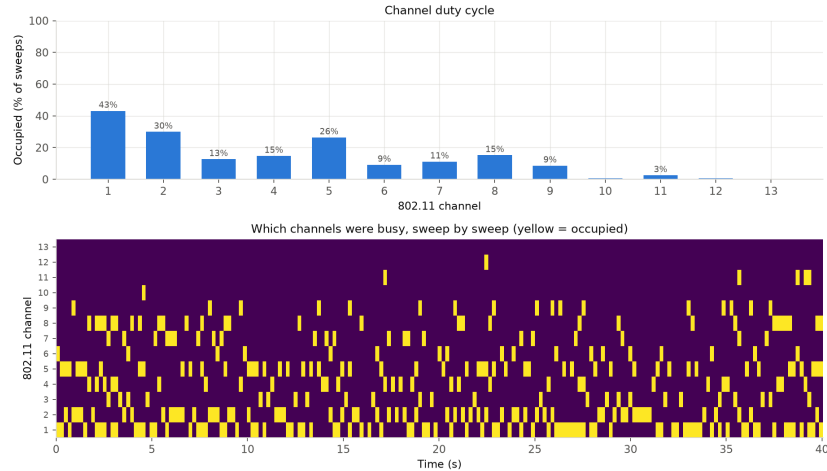
Figure 2 shows the same data over time as a heatmap. Wi-Fi frames show up as short horizontal streaks, and Bluetooth as small dashes above 2470 MHz. This confirms the activity is bursty and keeps changing from sweep to sweep. Figure 3 shows the occupancy: a bar chart of how often each channel was busy, and a channel-by-time grid.



Measured 2.4 GHz power spectrum over the whole band (mean and peak hold). The lower channels are busy; the region above channel 11 is nearly empty.



The band over time. Wi-Fi frames appear as horizontal streaks and Bluetooth hops as short dashes.



How often each channel was busy (top) and a channel-by-time grid (bottom).

We removed two things that come from the radio itself, not the air: the local-oscillator leak at 0 Hz and a spur at one eighth of the sample rate. Both showed up as fixed vertical lines. For the noise floor we measured from the low part of the spectrum instead of the median, because when most of the band is busy the median sits inside a signal and reads too high.

The most important check: we compared our result with the computer's own Wi-Fi scan of the same room, shown in Table II. They matched. Channels 1 (43%) and 2 (30%) were the busiest, which is where the scan lists the most networks, and the top channels were empty. Channel 5 read higher because a 40 MHz network on channel 1 spills into it.

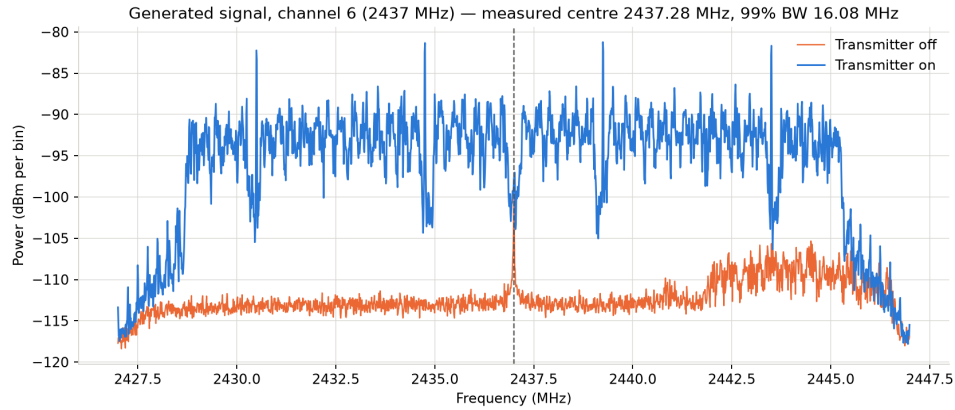
Table 2: Measured occupancy vs. independent Wi-Fi scan.

Channel	macOS scan	Measured %
1	3 (one 40 MHz)	43
2	7	30
5	- (ch1 40 MHz bond)	26
7-9	1 each	11-15
11-13	none	<=3

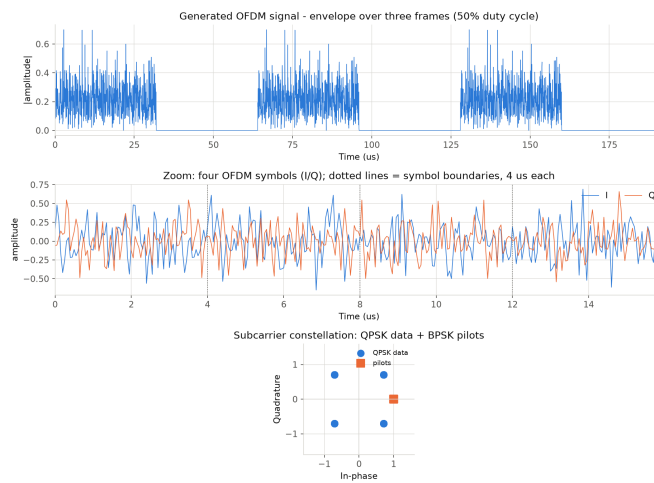
4. Part 2 - Wi-Fi-like Signal Generation

We checked the signal was really on the air by receiving it on the same radio while it transmitted. When we raised the transmit power, the received power went up by about the same amount, which only happens with real radio output.

Figure 4 shows the measured spectrum of our signal on channel 6. The centre came out at 2437.28 MHz (against 2437 nominal) and the width at 16.08 MHz inside the 20 MHz channel, close to the 16.45 MHz from theory and the ~16.6 MHz of real Wi-Fi. You can see the gap in the middle (the off sub-carrier) and the four pilot tones, which confirm the OFDM shape. Figure 5 shows the same signal in time: the on-off bursts, the noise-like I and Q samples over four symbols, and the QPSK data points with the pilots.



Measured spectrum of the generated signal on channel 6. The gap in the middle and the four pilot tones confirm the OFDM shape; measured centre 2437.28 MHz, width 16.08 MHz.



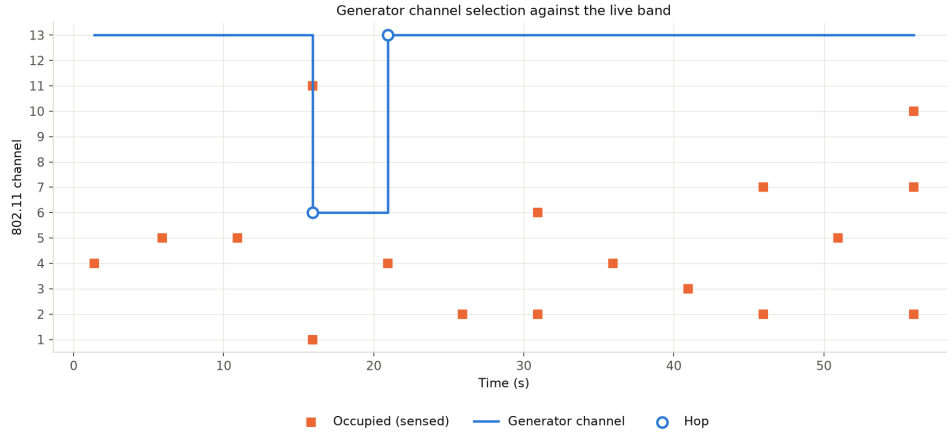
The generated signal in time: the on-off bursts (top), the I and Q samples over four symbols (middle), and the QPSK data with pilot tones (bottom).

5. Dynamic Channel Hopping

The generator picks its channel with a simple rule: among channels 1 to 13, choose the one whose nearest busy signal is furthest away, and if there is a tie, lean toward the middle of the band. Channel 14 is never used because it is not allowed here. With only channel 1 busy it goes to the far end; with both ends busy it sits in the middle of the gap.

A plain 'furthest away' rule is jumpy. With channels 1, 6 and 11 busy, channels 4, 8 and 13 are all the same distance away, so a little noise flips the winner and the generator hops every time for no reason. We added a margin: a new channel has to be clearly better (by 3 MHz) before it moves. We also decide from several sweeps, not one, so a single bad reading cannot push it onto a busy channel.

Figure 6 shows a 56-second run: the channel it chose against the busy channels it sensed. It settles on a clear channel and hops when the band changes, and it reacts in about 1.0 second. This is a trade-off: a shorter look reacts faster but misses weak, bursty signals; a longer look sees more but reacts slower.



The channel the generator chose against the busy channels it sensed.

6. Effect on the Active Wi-Fi Link

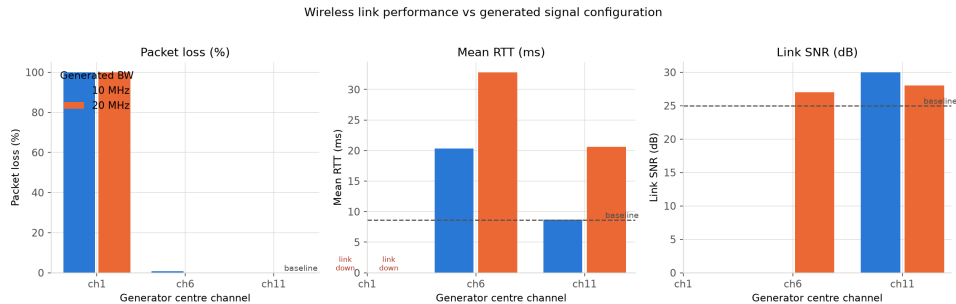
Last, we measured the effect on a real Wi-Fi link. The computer's Wi-Fi was on channel 1. We ran the generator at full power with the antenna near the laptop, and for each setting we recorded the round-trip time and packet loss to the router. Each run was compared with a 'transmitter off' reading taken just before it, to cancel the band's own drift.

The result is clear (Table III and Figure 7). When the generator was on channel 1 - the same channel as the link - the link lost 100% of packets and went down completely, in every repeat and at both widths. When we moved it 25 MHz away to channel 6, or 50 MHz to channel 11, the same signal caused no measurable loss.

Two things come out of this. First, what matters is being on the same channel, not the width, because 10 MHz and 20 MHz behaved the same at every offset. Second, this is exactly why the hopping in Part 2 helps: keeping distance from the busy channels is what protects a link.

Table 3: Link performance vs. generator channel.

Configuration	Offset	Packet loss	Link
Channel 1, 10 & 20 MHz	0 MHz	100% (6/6)	Down
Channel 6, 10 & 20 MHz	25 MHz	0 - 0.3%	Normal
Channel 11, 10 & 20 MHz	50 MHz	0%	Normal



Wi-Fi link vs generator setting. On channel 1 (the link's own channel) it is down; 25 and 50 MHz away it is unaffected.

7. Discussion

These results link back to the course. The busy, overlapping channels in Part 1 come from Wi-Fi's listen-before-talk rule and from the 5 MHz spacing against a 20 MHz channel width, which is why one access point shows up on several channel labels. The total packet loss in the last test is the interference-limited case: once another signal fills the same channel, Wi-Fi's carrier sensing never sees it free, so the link cannot send.

A few honest limits. The power values are relative, not calibrated against a lab reference, so the shapes and differences are right but the exact dBm numbers are not. We had one radio, so the two parts were measured separately, not together. And at low power we could not see any effect over the busy band's own changes; the clear result needed full power and a short antenna distance.

8. Conclusion

We built and measured a full 2.4 GHz sensing and interference system on the ADALM-PLUTO. The sensor finds the busy Wi-Fi channels across the band and tracks them over time, and we checked it against the computer's own scan. The generator makes a correct Wi-Fi-like signal (measured at 2437.28 MHz and 16.08 MHz) and hops away from busy channels in about 1.0 second. The interference test showed that a signal on the same channel takes a Wi-Fi link down completely, while the same signal one or two channels away does nothing. This shows, end to end, why choosing a clear channel matters.

9. References

- [1] IEEE Std 802.11, Wireless LAN Medium Access Control (MAC) and Physical Layer (PHY) Specifications, IEEE.
- [2] Analog Devices, ADALM-PLUTO SDR Active Learning Module, User Guide.
- [3] P. D. Welch, "The use of fast Fourier transform for the estimation of power spectra," IEEE Trans. Audio Electroacoustics, vol. 15, no. 2, 1967.
- [4] A. Goldsmith, Wireless Communications. Cambridge University Press, 2005.
- [5] T. S. Rappaport, Wireless Communications: Principles and Practice, 2nd ed. Prentice Hall, 2002.